PQC Migration (1) — Resource Estimates into Judging Criteria

Blockchain Governance Initiative Network

IKP session 1-4 · Workshop W2 of the PQC Migration Prize series

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BGIN Block #15 | Georgetown University Capitol Campus, Washington, D.C. | 15 October 2026, Room B, 09:50–11:20 | hybrid

Picks up from the opening keynote (Alex Pruden, Project Eleven, 09:20) and continues in session 1-7 at 11:30 · facts follow bgin-global/pqc-agility-competition, draft v0 · ecosystem-neutral: no chain and no scheme is endorsed

This Session — What It Is For

First of two IKP working sessions on the PQC Migration Prize (hosted by METI/NEDO; BGIN coordinates and runs the neutral testbed). From the keynote to the competition: crypto agility as the thing being evaluated — deployment, evaluation, testbeds and migration governance, not new primitives.

1. Tour of the public competition repository: governance, outcomes O1–O6, the 22-metric register (draft v0), the scheme register and the harness specification
2. Resource estimates and the ECDSA.fail community: what the benchmark numbers mean for migration timelines and for judging criteria
3. Open question: is the headline quantity total overhead or migration speed?
4. Open question: who must upgrade first — node operators or users — and how the metrics reflect that burden split
5. Open question: scoring the legitimate-owner-versus-attacker race during a migration window
6. Tagging each metric by the attack class it defends against — on-spend, at-rest, on-setup (the GDC26 taxonomy)

🔗 **Takeaway:** Leave with directions per metric for the v0 → v1 pass that session 1-7 consolidates, and owners for the three open questions above.

The Job Is Not to Pick Another Primitive

First PQC standards are in place (ML-KEM, ML-DSA, SLH-DSA; FN-DSA in draft; SP 800-208 stateful hash-based). Japan's PQC Migration Prize is hosted by METI/NEDO; BGIN coordinates internationally and runs the neutral evaluation testbed. It is not a BGIN contest, not a new-algorithm contest, and not chain endorsement.

What still has no agreed playbook for public ledgers, wallets, and credentials:

- How to stage a hybrid and then a cutover when every node must agree on validity rules.
- How to keep verifying old signatures for as long as funds and archives remain live.
- How to rotate keys and addresses without stranding dormant accounts.
- How to tell a genuine PQ-ready stack from a screenshot.

🔗 **Takeaway:** Treat PQC as a multi-year systems and governance program, not a weekend library swap. GDC26 named the gaps; Block 15 writes the criteria.

Session 1-4 — The Numbers on the Table

Migration urgency is a function of resource estimates. This room holds the estimate and the architecture in one place; policy rooms usually hear them from different speakers.

Source	What it says
Google Quantum AI et al., *PRX Quantum* 7, 031001 (Aug 2026)	secp256k1 ECDLP at about 1,200 logical qubits and 90M Toffoli gates, or 1,450 / 70M; attacks sorted **on-spend** / **at-rest** / **on-setup**; mitigations ranked by on-chain consequence
ECDSA.fail community benchmark	The continuing open resource-estimation benchmark: solvers and agents under public notes, re-run by a validator, promoted by measured improvement
Project Eleven, *Key recovery from state reuse* (Sep 2026)	The stateful hash-based failure is demonstrated, not theoretical: signing twice from one LMS / XMSS state exposes the key

🔗 **Takeaway:** The estimates set the clock. No calendar marketing of a single Q-day — threat narratives are scenarios, and the register scores which keys are already public.

On-Spend, At-Rest, On-Setup — Timing Is Not One Number

On-spend

The public key is revealed only when funds move. The race is between the legitimate owner and an attacker who sees the broadcast — a fast-clock CRQC during confirmation, not an unbounded offline grind.

At-rest

The public key is already on-chain. Any CRQC, at any later date, can grind offline against the stored verification key and forge a spend.

On-setup

A universal classical exploit manufactured once by a one-off quantum computation, before any attack — the case that breaks the “we will see it coming” assumption.

Address design, key reuse, and wallet defaults therefore change the threat model. Metric open question 6 asks, for each of the 22 metrics, which of the three classes it defends against. The same split returns in the Offline Key Management session (2-8): cold storage is at-rest by definition.

🔗 **Takeaway:** Resource estimates only become policy-useful when they are tied to which keys are already public.

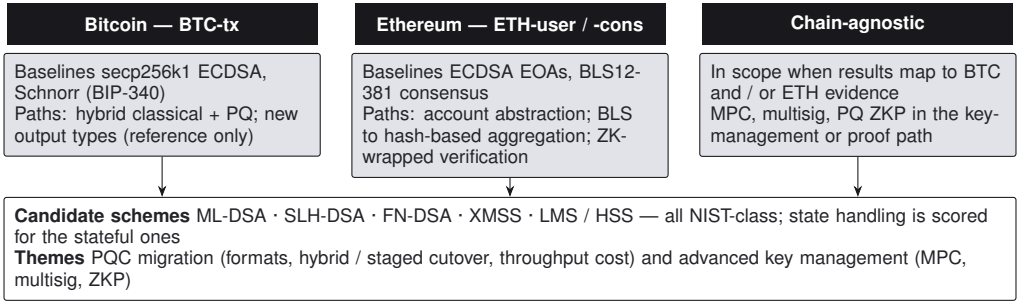
How an Estimate Becomes a Criterion

The register scores substantiation, not fear. Three rows carry the resource-estimate work; the rest are performance and agility.

- **M-19 Security review score** — checklist: assumptions, parameter set, side channels, stateful-key handling; seeds from the PQC Audit Index, to be adopted at W2.
- **M-22 PQC-readiness claim substantiation** — checklist against the *"PQC-ready chain" definition*, the December (W4) deliverable; scores a stack against a published definition rather than a claim.
- **M-20 Governance fit** — transition-period design, dormant / vulnerable asset treatment, the owner-versus-attacker race: a structured narrative, discussion input, not a policy decree.
- **Gate G1** — the primitive is NIST-standardized or NIST-selected, or a committee-ruled hybrid / stateful hash-based scheme. No new primitives, ever.

🔗 **Takeaway:** An estimate enters the rubric as a declared assumption with error bars, scored under M-19 and M-22 — never as a date.

Three Lanes and What Gets Evaluated



Comparative scoring is anchored on Bitcoin and Ethereum. Every performance number is a ratio against the classical baseline on the same node, reproduced on *k* nodes in at least two regions (gate G3).

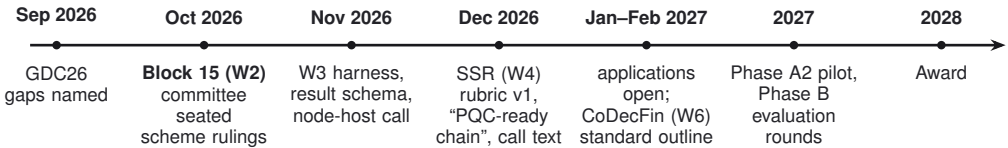
🔗 **Takeaway:** Japan funds the prize. BGIN turns shared evidence into standards-ready artifacts. NIST remains the reference for which primitives are in scope.

Session 1-4 — Discussion Items

1. What would make a **resource estimate** usable for an evaluator — assumptions, error bars, and what must not be collapsed into a Q-day date? What language does the rubric refuse (single Q-day, vendor screenshots)?
2. Which **objects** move first: at-rest keys, on-spend paths, credentials, the proof and key-management path (PQ ZKP, MPC)? Open question 6 needs an owner.
3. Is the headline quantity **total overhead** or **migration speed**? Who upgrades first, and how does M-12 record the burden split?
4. Should the ECDSA.fail benchmark's **public-note discipline** (exact model and harness named, re-run by a validator) be the admission rule for agent-produced numbers in the record? (Decision 0001.)
5. What does the **"PQC-ready chain" definition** have to contain for M-22 to be scorable by December?

Everything raised here is consolidated into the metric register v1 in session 1-7 at 11:30, where the scheme-scope rulings and the committee seating are taken.

From This Room to the 2028 Award



Competition / evaluation → BGIN Standard → ISO/TC 307 → Japan’s crypto-asset security criteria

Dates are indicative until the host confirms the call. SSR 2026 PQC-for-blockchains workshop: Sun 13 Dec, Baltimore; peer-reviewed papers there are intended to be reflected in the competition specification.

🔗 **Takeaway:** This morning can still change what the call asks teams to prove. SSR on 13 December is the next drafting stop.

How to Engage

- **Evaluation committee** — draft call text, metrics and rubrics in Phase A; nominate via the issue template, with a conflict-of-interest disclosure.
- **Technical advisor** — Bitcoin / Ethereum / measurement input without winner votes.
- **Metric or scheme proposal** — an issue, then a pull request to the register. Every commit is signed and signed-off; a PR is the record.
- **SSR workshop, 13 Dec** — accepted papers feed the competition specification.
- **Next**: session 1-7 at 11:30, same room — agility as a governance property, the NIST path, the W2 decisions.

This session
bgin-global.org/events/20261015-block15/sessions/1-4

Repository
github.com/bgin-global/pqc-agility-competition

Discourse category
bgin.discourse.group/c/post-quantum-crypto-agility-competition/44

🔗 **Takeaway:** Discussion on Discourse; the repository is the record. Nothing is final until the committee ratifies it and the host publishes the call.

An estimate enters the rubric as a declared assumption with error bars — never as a date.

The hard problem is deployment, evaluation, and migration governance.
Not selecting a new primitive. Not a Q-day. Not an endorsement.

BGIN IKP Working Group · Post-Quantum Crypto Agility Competition category
on the BGIN Discourse · github.com/bgin-global/pqc-agility-competition



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